Types of chain

a. Metric chain: Metric chains of length 5 m, 10 m, 20 m, 30 m are in use. Every metre is divided into 5 links each of length 0.2 m.

The length of the link is the distance between the centre of the two consecutive middle links. Small brass rings are provided at every metre and tallies at every 5 m length.

b. Non metric chain: The following are the non – metric chain in which the unit of measurement is in foot.

i. Engineer's chain: Engineer's chain is 100 feet long and is divided into 100 links, each link is being one foot long.

ii. Gunter's chain (or) surveyor's chain:

Gunter chain is 66 feet long and is divided into 100 links, each link is being 0.66 feet long.

iii. Revenue chain: Revenue chain is 33 feet long and is divided into 16 links.

Note: The above non – metric chains are not in use nowadays. They have become obsolete.

c. Steel bands: It is a steel ribbon with a brass swivel handle at each end. It is 3 to 16 mm wide, 20 to 30 m long and 0.2 to 0.6 mm thick. It is light weight and easy to handle

2. Tapes

Tapes are made of different materials.

The following are some of the tapes commonly used:

a. Linen (or) cloth tape

b. Glass fibre tape

c. Metallic tape

d. Steel tape

e. Invar tape



■ Glass fiber tape

It is made of glass fiber and PVC. It is flexible, strong and durable. It does not stretch or shrink due to change in temperature and dampness. It is available in varieties of lengths.

■ Steel tape

It is made of steel (or) stainless steel and used for accurate measurements. It is available in lengths of 5 m, 10 m, 20 m and 50 m.



Stephen Crane is the Inventor of digital tape.



3. Arrows

These are made of hardened and tempered steel wires of 4 mm diameter and length of 400 mm. These are pointed at one end and other end is bent to a circle of diameter 50 mm.

Arrows are used to mark the points on the ground or end of chain lengths and to note the number of chains measured during the chaining process.

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